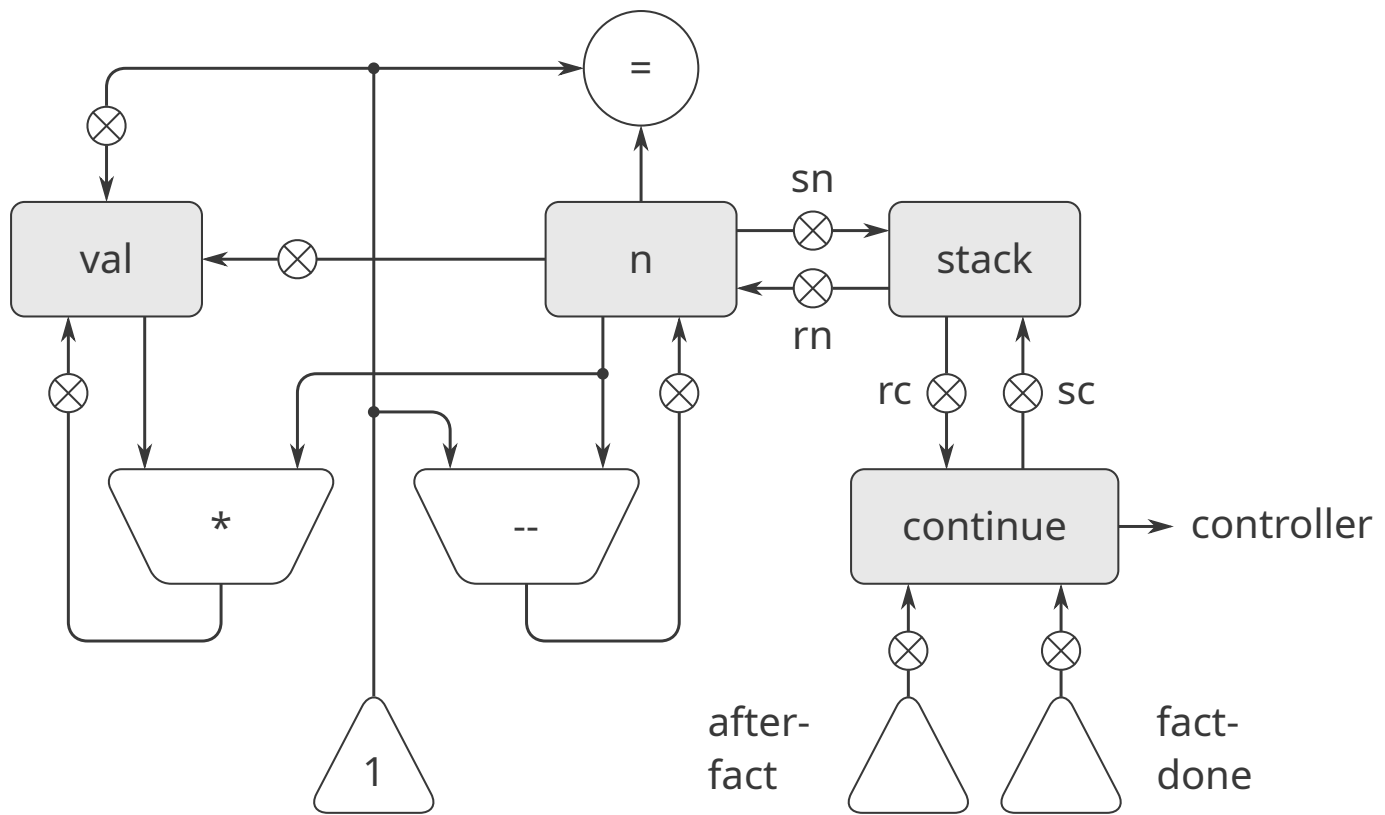




(controller

(assign continue (label fact-done)) ; set up final return address

fact-loop

(test (op =) (reg n) (const 1))

(branch (label base-case))

(save continue) ; Set up for the recursive call

(save n) ; by saving n and continue.

(assign n (op -) (reg n) (const 1)) ; Set up continue so that the

(assign continue (label after-fact)) ; computation will continue

(goto (label fact-loop)) ; at after-fact when the

after-fact ; subroutine returns.

(restore n)

(restore continue)

(assign val (op *) (reg n) (reg val)) ; val now contains $n(n - 1)!$

(goto (reg continue)) ; return to caller

base-case

(assign val (const 1)) ; base case: $1! = 1$

(goto (reg continue)) ; return to caller

fact-done)